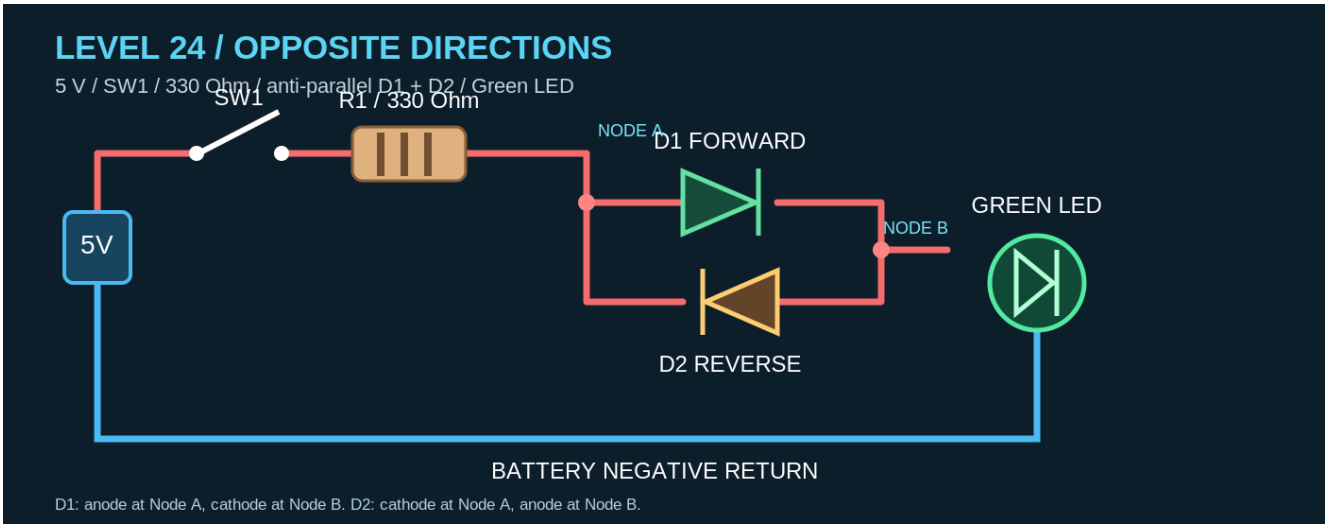
PROJECT SPARK — LEVEL 24 CREATION PROCESS

Opposite Directions — Diode Challenge

Game: Project Spark | **Level:** level_24 | **Circuit:** 5 V, 330 Ω, two opposing 1N4007 diodes, green LED | **Targets:** 6/6

This guide uses the uploaded and already prepared Level_24_Opposite_Diodes_Replacement.txt. It assumes your corrected resistance-based SparkDiodeSolverDefinition, diode Smart Slot category fix, and established level loader remain in place. No global solver changes are needed merely to assemble this level.

1 — Understand the required circuit



The diode pair is **anti-parallel**, not in series. The forward diode D1 carries current from the resistor to the green LED. The reverse diode D2 bridges the same two electrical nodes in the opposite direction; it must remain blocked.

```
BATTERY (+5 V) -- SW1 -- R1 330 Ω --+-- D1 ANODE ->|-- D1 CATHODE --+-- LED ANODE ->|-- LED CATHODE --
\BATTERY (-)
                                    |                                |
                                    +-- D2 CATHODE --|<- D2 ANODE ---+
```

Important: Both diode branches share exactly two electrical nodes, but D2 connects to them *with reversed polarity*. The two branches are not additional power supplies. Do not attach a bypass wire directly from the resistor output to the LED anode; that would bypass both diodes.

2 — Create the Level 24 prefab

1. Back up your completed Level 23 prefab and Level 23 definition.

2. Duplicate Level_23_Layout into Level_24_Layout; make the duplicate an independent prefab asset.

3. Rename the duplicate's level layout identity to level_24 in the relevant layout component/loader catalog.

4. Retain Power_Sources/PowerSupply_01, the existing power switch, workbench, fixed green LED, and the physical resistor slot.

5. Keep **two** diode workstations, renumbered Slot_02 and Slot_03. Remove the unused third diode workstation (former Slot_04) and any old runtime binding declaration for it.

6. Make sure the layout loader catalog includes the Level 24 prefab and the gameplay manager has a Level 24 entry. Existing progression may keep Level 24 locked until Level 23 completion.

7. Save the **original prefab** outside Play Mode. Do not make persistent changes to Level_24_Layout(Clone).

Recommended hierarchy:

```
Level_24_Layout
■■■ Power_Sources
```

```
■   ■■■ PowerSupply_01
■■■ Switch
■   ■■■ Power_Switch
■■■ Slot_01
■   ■■■ Resistor_330 Ω
■■■ Slot_02
■   ■■■ Diode_1n4007_01   (D1: forward)
■■■ Slot_03
■   ■■■ Diode_1n4007_02   (D2: reverse)
■■■ LED
    ■■■ led                (fixed SparkLED)
```

3 — Configure each SparkShelfSlot (Smart Runtime Bindings)

On the original prefab, enable Enable Smart Binding for all three workstation slots and confirm Category and Kind = Workstation:

Slot ID	Category	Runtime Component ID	Runtime Terminal A ID	Runtime Terminal B ID
Slot_01	Resistor	R_LIMIT_Component	R_LIMIT_Input	R_LIMIT_Output
Slot_02	Diode	D_FORWARD_Component	D_FORWARD_Anode	D_FORWARD_Cathode
Slot_03	Diode	D_REVERSE_Component	D_REVERSE_Anode	D_REVERSE_Cathode

Physical terminal object names inside both diode prefabs are Diode_Anode and Diode_Cathode (the same physical names can occur in both diode prefab instances). The resistor prefab uses Register_Input and Register_Output.

In SparkLevelWorkstationBindingBridge, keep **Use Smart Slots = ON**. Smart Slot diode matching must use suffixes _Anode and _Cathode rather than _Input/_Output. The smart slot's IDs take priority over legacy workstation rules. Avoid duplicate IDs or duplicate slot IDs. Make sure each diode's SparkDiode.AnodeTerminal and .CathodeTerminal are assigned to the matching terminal components.

4 — Declare runtime IDs in SparkLevelSceneBindings

Under **Runtime Component IDs**, declare exactly:

- R_LIMIT_Component

- D_FORWARD_Component

- D_REVERSE_Component

Under **Runtime Terminal IDs**, declare exactly:

- R_LIMIT_Input, R_LIMIT_Output

- D_FORWARD_Anode, D_FORWARD_Cathode

- D_REVERSE_Anode, D_REVERSE_Cathode

Keep the existing **fixed** bindings for PowerSupply_01, Battery_positive, Battery_negative, and the green LED. The imported definition uses fixed LED component led, terminal IDs LED_Anode and LED_Cathode. Ensure your prefab uses those exact IDs (do not silently replace with LED_INDICATOR_*). Do not declare the same alias as both fixed and dynamically owned.

5 — Correct 7 external wire connections

Use the **actual SparkTerminal objects** while drawing wires in game:

Wire	From	To
1	Battery_positive	Switch input
2	Switch output	Register_Input

Wire	From	To
3	Register_Output	D1 Diode_Anode
4	Register_Output	D2 Diode_Cathode
5	D1 Diode_Cathode	LED_Anode
6	D2 Diode_Anode	LED_Anode
7	LED_Cathode	Battery_negative

The resistor output and LED anode each require **two wire connections**. Check that SparkCircuitSystem and your connection rules permit multi-attachment to a terminal; if not, do not force this topology without adapting that policy. A wire from D2 cathode to the resistor output and from D2 anode to LED anode is a reverse branch; D2 should not illuminate or pass appreciable forward current.

6 — Import Level_24 definition

1. Select the original Level_24.asset in the Project window.

2. Import Level_24_Opposite_Diodes_Replacement.txt with your custom editable ScriptableObject importer.

3. Keep its existing GUID and source object references. Do not create an unrelated asset by pasting the JSON.

4. Confirm levelId = level_24, name Opposite Directions - Diode Challenge, requiredTargetCount = 6, requireClosedReturn = true, rejectShortCircuit = true, rejectTargetShort = true.

5. Inspect target IDs against the table below before testing.

Six imported objectives

Target ID	Type	Required binding	Requirement
level24_supply	VoltageRange	PowerSupply_01	4.8–5.2 V
level24_d1_forward	VoltageRange	D_FORWARD_Component; forward anode/cathode	0.45–1.0 V
level24_d2_blocked	CurrentRange	D_REVERSE_Component; reverse anode/cathode	0–0.00005 A
level24_led_on	LEDOn	led; LED_Anode, LED_Cathode	ON
level24_safe_current	CurrentRange	R_LIMIT_Component	0.005–0.008 A
level24_resistor_voltage	VoltageRange	R_LIMIT_Component	2.0–2.6 V

Critical evaluator validation: The reverse branch is expected to have zero current. Some evaluator implementations reject an entirely nonconducting target or fail to resolve its current. Test level24_d2_blocked independently. If it cannot pass at 0 A, add a purpose-built reverse-blocked predicate in the evaluator *only after confirming the evaluator's actual code*; don't falsely mark the level complete or change diode physics to make the target pass.

7 — Test order

1. Start Level 24, verify Active layout: True and all three workstations are registered.

2. Press **Synchronize AND REPORT**: all 3 component IDs and all 6 terminal IDs must show bound to the right instances.

3. Close SW1 and connect the exact seven wires.

4. Run a solve and check SolveNow success, source output, resistor, LED and both diode states.

5. Expected approximate values: supply 5 V; resistor ~2.3 V, ~6.9 mA; D1 +0.7 V, ~6.9 mA; D2 -0.7 V, approximately 0 A; LED ~2.0 V, lit. Values depend on your actual solved device models.

6. Verify Closed Return = YES, Source Short = NO, Target Short = NO.

7. Verify 6/6 targets individually; if only 5/6 passes, inspect the failed target before changing thresholds.

8. Save prefab and level definition outside Play Mode, retest starting from the chapter/level manager.

8 — Troubleshooting

Symptom	Likely next check
D_FORWARD_* / D_REVERSE_* UNBOUND	Slot category Diode; Smart Slots ON; binding IDs declared; bridge _Anode/_Cathode suffix matching
Both diodes pass current	Check D2 polarity and both branch endpoints; make sure D2 cathode connects to resistor output
Source current 0 A	Switch closed, connected return, diode orientation, solver convergence
Only one branch has voltage	Graph terminal registration and multi-attachment capability; inspect measured terminal voltage on both sides
D2 blocked, yet level 5/6	Verify evaluator accepts valid 0 A for CurrentRange and that D_REVERSE_Component resolves
Resistor too much/too little current	Check actual resistor is 330 Ω, supply is 5 V, LED model voltage, and no bypass wire
Logs flood Console	Remove temporary per-iteration diode logs and diagnostic scans after testing

9 — Drawing pages (optional visual teaching)

1. **Introduction:** Forward vs reverse bias and two opposing paths.

2. **Wiring:** Seven correctly numbered connections; show the two shared nodes.

3. **Operation:** D1 conducts; D2 blocks. Explain signed voltage across D2.

4. **Success:** Expected measurements and six objectives; actual completion must come from the level evaluator.

Preserve: Your working Level 21 transient driver, corrected AddForwardDrop() resistance stamp, current diode hysteresis/convergence fixes, and Smart Slot diode-category support. Do not reintroduce global continuous solving.